

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms two equations of motion for 1.8 and 1.2 kg mass, three terms, condone sign error. "Whole string" approach scores M0	AO3.4	M1	$1.8 \times 9.81 - T = 1.8 \times a$
	Obtains two correct equations	AO1.1b	A1	$T - 1.2 \times 9.81 = 1.2 \times a$
	Solves to find correct value for a 0.2g or AWRD 1.96	AO1.1b	A1	$a = 1.96 \text{ m s}^{-2}$
	Uses $s = ut + \frac{1}{2}at^2$ with <i>their</i> calculated a value	AO1.1a	M1	$1.5 = \frac{1}{2} \times 1.96 \times t^2$
	Calculates correct t value AWRT 1.24	AO1.1b	A1	$t = 1.24 \text{ seconds}$
(b)	Any valid assumption stated. Do not allow any comment already stated as an assumption in the question. Air resistance is permitted.	AO3.5b	E1	The string is long enough so that lighter mass does not reach the peg before the heavier mass hits the ground.
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model for maximum friction = μmg	AO3.3	B1	$F_{\max} = \mu mg$
	Makes an appropriate comparison	AO1.1a	M1	$= 0.85 \times 20 \times 9.8$ $= 166.6 \text{ N}$
	Explains clearly why crate remains stationary	AO2.4	E1	$150 < 166.6$ $\therefore$ crate does not move
(b)	Forms an equation by resolving vertically Condone one of sign error	AO3.1b	M1	$20g = R + 150\sin 15^\circ$ $R = 157.177 \text{ N}$

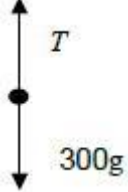
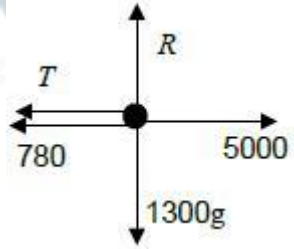
or cos error			$F_{\max} = \mu \times 157.177$ $= 133.6 \text{ N}$ $150\cos 15^\circ = 145 \text{ N}$ $145 > 133.6$ $\therefore$ crate begins to move
Obtains correct reaction force	AO1.1b	A1	
Uses maximum friction = μR With 'their' reaction force Must identify maximum or limiting friction	AO1.1b	B1F	
Compares $150\cos 15^\circ$ with 'their' maximum friction	AO1.1a	M1	
Explains, using their values, why the crate begins to move.	AO2.4	E1F	
			Total 8 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Forms equation of motion with four correct terms Condone sign error	AO3.4	M1	$300 - 140 - R = 482 \times 0.2$ $R = 63.6 \text{ N}$
	Obtains correct equation.	AO1.1b	A1	
	Obtains correct value of R .	AO1.1b	A1	
(ii)	Forms equation of motion with correct terms Condone sign error	AO1.1a	M1	$T - 63.6 = 72 \times 0.2$ $T = 78 \text{ N}$
	Obtains correct equation Follow through their R	AO1.1b	A1F	
	Obtains correct value of T	AO1.1b	A1	
(b)	States appropriate assumption NOT friction or air resistance	AO3.3	E1	Rope has no mass or is horizontal or is inextensible
(c) (i)	Forms equation of motion for skater using 'their' R Condone sign error	AO3.1b	M1	$-63.6 = 72a$ $a = -0.883... \text{ m s}^{-2}$ $u = 6 \text{ v} = 0 \text{ a} = -0.883$
	Finds correct acceleration for 'their' R	AO1.1b	A1F	
	Uses a suitable constant acceleration formula with	AO1.1a	M1	

	'their' a			
	Obtains s when $v = 0$ Or Obtains v or positive v^2 when $s = 20$	AO1.1b	A1F	$0 = 6^2 - 2 \times 0.883s$ $s = 20.4 \text{ m}$
	Explains that the skater hits buggy using correct values	AO3.2a	E1	$20.4 > 20$ Skater hits buggy
(ii)	Explains that the tension is removed from the buggy	AO2.4	E1	The rope is released so there is no tension acting on the buggy, so there is a higher resultant force. The driver will notice an increase in acceleration.
	Explains that the driver notices an increase in acceleration	AO2.4	E1	
Total 14 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws correct force diagram for crate from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(ii)	Draws correct force diagram for van from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(b)	Applies Newton's 2nd Law ($F = ma$) to the crate	AO3.4	M1	For crate $T - 300g = 300a$
	Applies Newton's 2nd Law ($F = ma$) to the van ($F = ma$ 'round the corner' scores 0)	AO3.4	M1	For van $5000 - T - 780 = 1300a$ $4220 - T = 1300a$
	Solve their simultaneous equations	AO1.1a	M1	$4220 - 300g = 1600a$

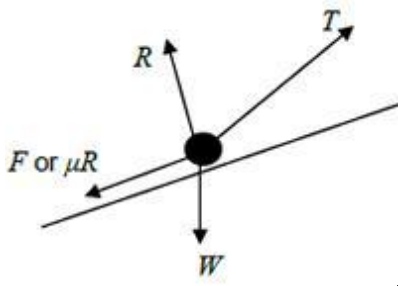
	Finds the value of a correctly AG	AO1.1b	A1	$a = 1280 \div 1600 = 0.80 \text{ m s}^{-2}$ (AG)
(c)	Uses $a = 0.80$ in either of their two equations in (b)	AO3.4	M1	$T = 300 \times 0.80 + 300g$
	Finds the correct value for T (condone omission of units) Possibly done in (b)	AO1.1b	A1	$= 3180$ $= 3200 \text{ (N) (2 sf)}$
(d)	Explains that the model could be refined by including air resistance	AO3.5c	E1	Resistance will increase with speed
Total 9 marks				

Q5.

Marking Instructions	AO	Marks	Typical Solution
Uses correct forces to form a moment equation (PI)	AO1.1a	M1	Take moments about C: $Mg \times 0.8 = 0.7 \times 24$
Obtains correct value	AO1.1b	A1	$M = 21$
Total 2 marks			

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40^\circ$ Resolving horizontally $F = 50\cos 40^\circ$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50\cos 40^\circ = \mu(8 \times 9.8 - 50\sin 40^\circ)$ $\mu = \frac{50\cos 40^\circ}{8 \times 9.8 - 50\sin 40^\circ}$
	Completes a rigorous	AO2.1	R1	$\mu = 0.8279660445 = 0.83$

	argument that results with correct μ AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips			(2sf) AG
(b) (i)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R , $8g\cos 5^\circ$ and $T\sin 40^\circ$ (or $T\cos 50^\circ$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8\cos 5^\circ - T\sin 40^\circ$ $= 78.1 - T\sin 40^\circ$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T\cos 40^\circ - 8 \times 9.8\sin 5^\circ - F = 8 \times 3$ $T\cos 40^\circ - 8 \times 9.8\sin 5^\circ - 0.827... \times (8 \times 9.8\cos 5^\circ - T\sin 40^\circ) = 24$
	Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \frac{24 - 8 \times 9.8\sin 5^\circ + 0.827... \times 8 \times 9.8\cos 5^\circ}{\cos 40^\circ + 0.827... \times \sin 40^\circ}$ $T = 73.55939193 = 74 \text{ (2 sf)}$
	Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
	Uses the friction model in the four term equation of motion where R is in the	AO3.1b	dM1	

form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1		
Solves for T	AO1.1b	A1
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F
Total 12 marks		

Q7.

(a)

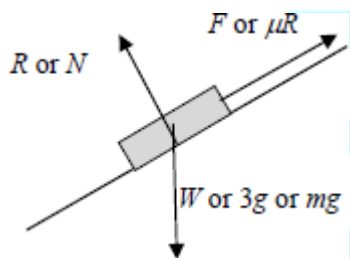


Diagram with exactly three forces showing arrow heads and labelled. If components are also shown they must use a different style e.g. dashed lines then they can be ignored. Friction must be up the slope.

B1

1

(b) $(R =) 3 \times 9.8 \cos 40^\circ = 22.5 \text{ N}$

M1: Resolving perpendicular to the slope. Can use $\sin 40^\circ$ or $\cos 50^\circ$ for method mark, with g or 9.8 .

A1: Correct normal reaction. Accept
 AWR T 22.5 (Note use of 9.81 still gives 22.5 N.)

M1A1

2

(c) $(F =) 0.2R = 4.50 \text{ N}$

M1: Use of $F = \mu R$.

A1F: Correct friction. Accept 4.5 N or AWR T 4.50.
 (Accept 4.51 N from the use of 9.81.)

Follow through incorrect normal reaction from part (b).

M1A1F

2

(d) $3a = 3 \times 9.8 \sin 40^\circ - 4.504$

M1: Three term equation of motion with correct terms, with $3a$, either component of weight and their answer to part (c) for F .

A1F: Equation of motion with correct terms and signs.

M1A1F

$a = 4.80 \text{ m s}^{-2}$

A1F: Correct acceleration. Accept 4.8 or AWRT 4.80.

(Note that using 9.81 still gives 4.80 m s^{-2}).

Follow through friction from part (c).

A1F

3

- (e) No air resistance force acting
 or
 No other forces acting on the box.
 or
 They (forces in the diagram) are the only forces that act.
OR
 No turning effect (due to forces).
 or
 Forces are concurrent. OE

B1: Correct assumption.

Ignore irrelevant comments

B1

1

EXAM PAPERS PRACTICE

[9]

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Q8.

(a) $5900 \times 0.2 = 2500 - 800 - R$

M1: Equation of motion for tractor and trailer as a single particle, with 2500, 800, R (which might be implied by seeing 1180 and 1700 or 1180 and 3300) and 5900×0.2 OE, with any signs.

A1: Correct equation.

M1A1

$(R =)2500 - 1180 - 800 = 520 \text{ N}$

A1: Correct R .

A1

If tension found first, do not award any marks until an

equation for R is obtained. Award M1 for
 $3500 \times 0.2 = \pm 2500 \pm R \pm 1280$.

3

(b) $T - 800 = 2400 \times 0.2$

M1: Equation for trailer with 2400 and 800.

A1: Correct equation.

M1A1

$(T =) 800 + 480 = 1280 \text{ N}$

A1: Correct tension.

A1

3

OR

$3500 \times 0.2 = 2500 - 520 - T$

M1: Equation for tractor with 3500, 2500 and 520.

A1F: Correct equation.

(M1A1F)

$(T =) 2500 - 700 - 520 = 1280 \text{ N}$

A1F: Correct tension.

(A1F)

(3)

Follow through incorrect R from part (a).

If the tension has been found in part (a) it only needs to be stated here.

(c) 1280 N

B1F: Same answer as part (b).

Do not accept – 1280.

B1F

1

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Q9.

(a) $3g - T = 3a$

M1: Three term equation of motion with $3g$ or 29.4 , T and $3a$.

A1: Correct equation.

M1A1

$T - g = a$

$$2g = 4a$$

M1: Three term equation of motion with g or 9.8 , T and a .

A1: Correct equation.

M1A1

$$a = \frac{g}{2} = 4.9 \text{ m s}^{-2}$$

A1: Correct final answer. Accept $\frac{g}{2}$

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 4.9, having seen -4.9 in their working. If the final answer is -4.9 don't award the final A1 mark.

Special Case:

Whole string method $2g = 4a$ and $a = \frac{2g}{4} = 4.9$ OE scores M1A1A1

5

(b) $v^2 = 0^2 + 2 \times 4.9 \times 0.4$

M1: Use of a constant acceleration equation to find v , with $u = 0$, their value for a from part (a) and $s = 0.4$ or 40 .

M1

$$v = \sqrt{3.92} = 1.98 \text{ m s}^{-1}$$

A1F: Correct speed. Follow through their acceleration from part (a).

Use $v = \sqrt{0.8a}$ for FT.

Accept 1.97.

If candidates use two equations, award no marks until they have an equation for v . (Note use of $t = 0.404$ or better required for A1)

A1F

2

(c) $0^2 = (\sqrt{3.92})^2 + 2 \times (-9.8) s$

M1: Use of a constant acceleration equation with $v = 0$, $a = \pm 9.8$ and their speed from (b).

M1

$$s = \frac{3.92}{2 \times 9.8} = 0.2 \text{ m}$$

A1: Correct distance.

A1

$$\text{Total} = 0.2 + 0.4 = 0.6 \text{ m}$$

A1: Correct total distance. Allow 60 cm from correct working.

Note

$$0^2 = \frac{(\sqrt{392})^2}{392} + 2 \times (-9.8)s$$

$$s = \frac{392}{2 \times 9.8} = 20$$

scores M1A0A0

A1

If candidates use two equations, award no marks until they have an equation for s . (Note use of $t = 0.202$ or better required for A marks)

3

- (d) The acceleration would be less, because the resultant force on each particle would be reduced.

B1: Less

'Slower acceleration' not acceptable

B1

B1: Appropriate reason.

Only award second B1 if they say acceleration is less.

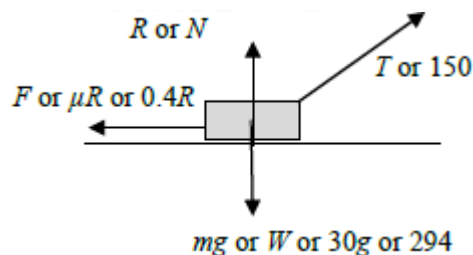
B1

2

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Q10.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.

B1: Diagram with one error or omission.

B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20^\circ = 30 \times 9.8$

$(R =) 30 \times 9.8 - 150 \sin 20^\circ$

$= 242.69 \dots$

M1: Resolving vertically to obtain a three term equation, with R , $150 \sin$ or $\cos(20^\circ$ or $70^\circ)$ and $30g$ oe.

A1: Correct equation. Allow g instead of 9.8 .

M1A1

$= 243 \text{ N (to 3sf)}$

A1: AG Correct final answer having seen either 2nd or 3rd or both line of solution.

A1

3

(iii) $(F =) 0.4 \times 242.7 = 97.1 \text{ N}$

M1: Use of $F = \mu R$ or $F \leq \mu R$.

A1: Correct final answer without an inequality. Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20^\circ - 97.08$

M1: Three term equation of motion with $30a$, $150 \sin$ or $\cos(20$ or $70^\circ)$ and their friction from (a)(iii). Condone incorrect signs.

A1: Correct equation.

M1A1

$$a = \frac{150 \cos 20^\circ - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a .

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWRT 1.46.

A1

4

(b) $R = 30 \times 9.8 - T \sin 20^\circ$

B1: Correct normal reaction in terms of T .

B1

$$F = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

B1: Correct friction in terms of T

B1

$$T \cos 20^\circ = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

*M1: Resolving tension horizontally and equating to F ,
provided that F is in terms of T .*

A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20^\circ + 0.4 \sin 20^\circ} = 109 \text{ N}$$

A1: Correct tension. AWR 109.

A1

5

(c) The same

B1: The same.

Use of $g = 9.81$ gives acceptable final answers.

B1

1

[17]

Q11.

(a) (i) Moments about Q

M1

$$2.2 \times 25g = T_p \times 4.2$$

$$T_p = 13.095 \times g$$

A1

$$T_p = 128 \text{ N}$$

A1

Resolving vertically

$$T_p + T_q = 25g \text{ or } 245$$

M1

$$T_Q = 117 \text{ N}$$

A1

OR

Moments about any point M1A1.

Moments about any other point M1.

$$T_P A1 ; T_Q A1$$

5

(ii) Weight of plank acts through its centre.

E1

1

(b) Resolve vertically

Could use T rather than T_P, T_Q

M1

$$T_P + T_Q = (25 + m)g = 2T_P$$

A1

Moments about B

M1

$$T_P \times 5 + T_Q \times 0.8 = 25g \times 3$$

$$(25 + m)g \times 2.9 = 25g \times 3$$

A1

$$2.9mg = 25g \times 0.1$$

$$29m = 25$$

M1

$$m = 0.862 \text{ or } \frac{25}{29}$$

A1

OR

Moments about Q

$$T_P \times 4.2 = 25g \times 2.2 - mg \times 0.8$$

$$\frac{1}{2} \times (25 + m)g \times 4.2$$

$$= 25g \times 2.2 - mg \times 0.8$$

$$2.9mg = 25g \times 0.1$$

$$29m = 25$$

OR

Moments about any point M1A1

Moments about any other point M1A1

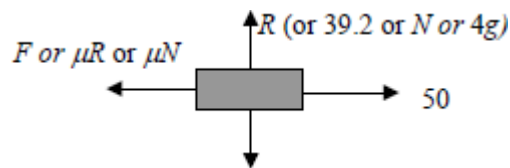
Solution M1A1

6

[12]

Q12.

(a)



mg or W or $4g$ or 39.2

B1: Correct force diagram with four forces with arrows and labels. Accept words eg friction instead of letters. Ignore negative signs in labels.

Do not accept 4 kg for the weight.

Award marks if forces are drawn on the diagram in the question.

B1

1

(b) 39.2 N

B1: Correct reaction force.

Accept 4g. Do not accept 39.

B1

1

(c) $50 - F = 4 \times 3$

M1: Three term equation of motion with the correct terms.

A1: Correct equation with correct signs.

M1A1

$$F = 38$$

A1: Correct friction.

A1

3

(d) $38 = \mu \times 39.2$

M1: Use of $F = \mu R$ with their answers to (b) and (c).

M1

$$\mu = \frac{38}{39.2} = 0.969$$

A1F: Correct μ based on their answers to (b) and (c).
Accept AWRT 0.969.

A1F

Note: $F = 12$ leads to 0.306 and award M1 A1F
Condone 0.97 or FT to 2sf
Condone use of inequalities.

2

- (e) Less friction, so a smaller coefficient of friction.

B1: Less friction.

B1

B1: Smaller μ .

B1

Note:

More friction anywhere scores B0 B0

Less friction, greater μ scores B1 B0

Smaller μ with no / inexact reason B0 B1

2

[9]

Q13.

- (a) $4720 - 3R = 2200 \times 1.6$

M1: Three term horizontal equation of motion with mass of 2200 kg and $3R$ (or $2R$ and R).

A1: All terms correct (4720 , $3R$ and 2200×1.6).

M1A1

A1: Correct signs.

A1

$$R = \frac{4720 - 3520}{3} = 400$$

A1: Correct value for R .

A1

OR

$$4720 - R - T = 1200 \times 1.6$$

$$T - 2R = 1000 \times 1.6$$

$$4720 - 3R = 3520$$

M1: Forming an equation for each body and adding to eliminate T .

A1: Two correct equations.

(M1A1)

A1: Correct equation in R .

(A1)

$$R = 400$$

A1: Correct value for R .

(A1)

4

(b) $T - 2 \times 400 = 1000 \times 1.6$

M1: Three term equation of motion for caravan with T , $2R$ and 1000×1.6 .

A1F: Correct equation, with their value for R from part (a).

M1A1F

$$T = 800 + 1600 = 2400 \text{ N}$$

A1F: Correct tension. Follow through from part (a) using $T = 1600 + 2R$

A1F

OR

$$4720 - T - 400 = 1200 \times 1.6$$

M1: Four term equation of motion for car with 4720, T , R and 1200×1.6 .

(M1)

$$T = 4720 - 400 - 1920 = 2400 \text{ N}$$

A1F: Correct equation, with their value for R from part (a)

(A1F)

A1F: Correct tension. Follow through from part (a) using $T = 2800 - R$

Note: do not follow through if a negative value is used for R .

(A1F)

3

[7]

Q14.

(a) $20 \cos \theta = 10$

M1: Resolving horizontally. Accept $\sin \theta$ or $\cos \theta$ with the 20.

$$\cos \theta = \frac{1}{2}$$

A1: Correct equation.

M1A1

$$\theta = 60^\circ$$

A1: Correct angle.

Accept $\frac{\pi}{3}$ or 1.05 (radians).

Allow 59.9 or better if they find W first

A1

3

(b) $(W =)20 \sin 60^\circ$

$$= 17.3 \text{ N}$$

Or

$$(W =)\sqrt{20^2 - 10^2} = 17.3 \text{ N}$$

M1: Resolving vertically. Accept $\sin \theta$ or $\cos \theta$ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

A1: Correct weight CSO

or

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

A1: Correct weight CSO

Accept $10\sqrt{3}$ or AWRT 17.3

M1A1F

2

(c) $m = \frac{20 \sin 60^\circ}{9.8} = 1.77 \text{ kg}$

M1: Their answer to part (b) divided by 9.8.

A1F: Correct mass. Follow through their answer to part (b).

Accept 1.76 or 1.8.

Accept 2 sig figs in follow through.

Note: Using $g = 9.81$ gives the answer

1.77, also accept 1.76.

M1A1F

2

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Q15.

$$20g - T = 20a$$

M1: Three term equation of motion for one particle.

M1

$$T - 5g = 5a$$

$$15g = 25a$$

M1: Three term equation of motion for the other particle.

A1: Both equations correct.

M1A1

$$a = \frac{15g}{25} = 5.88 \text{ ms}^{-2}$$

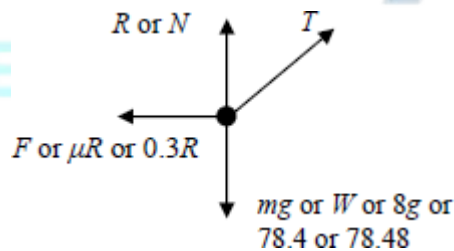
A1: Correct acceleration from correct working.

A1

[4]

Q16.

(a)



B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or $0.3R$ for F .

B1

1

(b) $R + 40 \sin \theta = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R , $T \sin$ or $\cos(30^\circ$ or $60^\circ)$ and $8g$ oe.

A1: Correct equation

M1A1

$$R = 78.4 - 40 \sin \theta$$

A1: Correct expression for R .

Accept $(R=)8g - T \sin 30^\circ$

Note if using $g = 9.81$ accept

$$R = 78.48 - 0.5T \text{ or } R = 78.5 - 0.5T$$

A1

3

(c) $8a = 40 \cos \theta - 0.3(78.4 - 40 \sin \theta)$

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

$$8a = -23.52 + 40 \cos \theta + 12 \sin \theta$$

M1: Use of equation of motion with $40 \sin \theta$ or $40 \cos \theta$ and $8a$ and a friction term.

M1

$$p = -2.94$$

$$q = 8$$

$$r = 1.5$$

A1: One correct value.

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A1

A1: All three values correct.

A1

5

[9]

Q17.

(a) $20 \cos \theta = 10$

M1: Resolving horizontally. Accept $\sin \theta$ or $\cos \theta$ with the 20.

A1: Correct equation.

M1A1

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

A1: Correct angle.

Accept $\frac{\pi}{3}$ or 1.05 (radians).

Allow 59.9 or better if they find W first

A1

3

(b) ($W =$) $20 \sin 60^\circ$

M1: Resolving vertically. Accept $\sin \theta$ or $\cos \theta$ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

M1

$$= 17.3 \text{ N}$$

A1: Correct weight CSO

A1

Or

$$(W =) \sqrt{20^2 - 10^2} = 17.3 \text{ N}$$

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

(M1)

A1: Correct weight CSO

Accept $10\sqrt{3}$ or AWR 17.3

(A1)

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2

(c) $m = \frac{20 \sin 60^\circ}{9.8}$

M1: Their answer to part (b) divided by 9.8.

M1

$$= 1.77 \text{ kg}$$

A1F: Correct mass. Follow through their answer to part (b).

Accept 1.76 or 1.8.

Accept 2 sig figs in follow through.

Note: Using $g = 9.81$ gives the answer 1.77, also accept 1.76.

A1F

Q18.

(a) $18g - T = 18a$

M1: Three term equation of motion for the 18 kg particle.

A1: Correct equation of motion for the 18 kg particle.

(Accept $T - 18g = 18a$)

M1A1

$T = 12a$

B1: Equation of motion for the block that has signs consistent with the first equation.

B1

$18g - 12a = 18a$

$a = \frac{18g}{30} = 5.88 \text{ ms}^{-2}$

*A1: Correct acceleration from correct work. Accept $\frac{3g}{5}$
Do not penalise consistent use of negative acceleration, provided final answer positive.*

Special Case:

*Whole String Method $18g = 30a$ and $a = \frac{18g}{30} = 5.88$
OE M1A1*

Note using $g = 9.81$ gives 5.89, also accept 5.88.

A1

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4

(b) (i) $18g - T = 18 \times 3$

M1: Three term equation of motion for the 18 kg particle with $a = 3$ seen.

A1: Correct equation.

M1A1

$T = 18g - 18 \times 3 = 122(.4)\text{N}$

A1: Correct tension. Accept 122.4.

Note using $g = 9.81$ gives 123, also accept 122.

A1

3

(ii) $(R = 12 \times 9.8 = 117.6 \text{ N} \Rightarrow) 118 \text{ N (to 3sf)}$

*B1: Correct normal reaction. Accept 117 and 117.6.
 Final answer must be positive.
 Do not accept 12g.
 Note using $g = 9.81$ gives 118, also accept 117.*

B1

1

(iii) $122.4 - F = 12 \times 3$

*M1: Three term equation of motion for the block,
 containing their tension, F and 12×3 .*

*A1F: Correct equation. Follow through T from
 part (b)(i).*

M1A1F

$F = 86.4$

A1F: Candidate's T minus 36.

A1F

$86.4 = \mu \times 117.6$

*dM1: Use of $F = \mu R$ with AWRT 117 or 118 for R and
 the candidate's value of F provided positive.*

dM1

$\mu = \frac{86.4}{117.6} = 0.735$

*A1: Correct μ . Accept anything between 0.728 and
 0.739 inclusive. Allow 0.73 and 0.74.*

A1

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5

- (c) No air resistance or no other forces

B1: One assumption from list

B1

Horizontal String

Block is a particle or **they** are particles

B1: For another assumption from list.

Do not penalise assumptions not in the list.

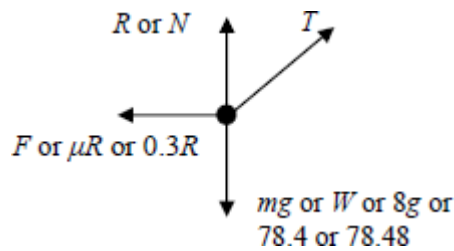
B1

2

[15]

Q19.

(a)



B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note: Accept μR or $0.3R$ for F .

B1

1

(b) $R + T \sin 30^\circ = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R , $T \sin$ or $\cos(30^\circ$ or $60^\circ)$ and $8g$ oe.

A1: Correct equation

M1A1

$(R =) 78.4 - T \sin 30^\circ$

$(R =) 78.4 - 0.5 T$

A1: Correct expression for R .

Accept $(R =) 8g - T \sin 30^\circ$

Note if using $g = 9.81$ accept

$R = 78.48 - 0.5T$ or $R = 78.5 - 0.5T$

A1

3

(c) $T \cos 30^\circ - F = 8 \times 0.05$

M1: Horizontal equation of motion with F , $T \sin$ or $\cos(30^\circ$ or $60^\circ)$ and 8×0.05 oe.

A1: Correct equation.

M1A1

$F = 0.3(78.4 - \sin 30^\circ)$

M1: Using $F = 0.3R$ with their R from part (b), provided it includes a term in T .

A1: Correct expression for friction.

M1A1

$$T \cos 30^\circ - 0.3(78.4 - T \sin 30^\circ) = 0.4$$

$$T = \frac{23.52 + 0.4}{\cos 30^\circ + 0.3 \sin 30^\circ} = 23.5\text{N}$$

dM1: Solving for T . Must see $(\cos 30^\circ \pm 0.3 \sin 30^\circ)$ or similar in the denominator. (Dependent on both previous M marks.)

A1: Correct T . Accept 23.6 or AWRT 23.5

dM1A1

6

Or

$$T \cos 30^\circ - F = 8 \times 0.05$$

M1: Horizontal equation of motion with F , $T \sin$ or $\cos(30^\circ \text{ or } 60^\circ)$ and 8×0.05 oe.

A1: Correct equation.

(M1A1)

$$T \cos 30^\circ - 0.3R = 8 \times 0.05$$

M1: Using $F = 0.3R$

A1: Two correct equations involving only T and R .

(M1A1)

$$R + T \sin 30^\circ = 8 \times 9.8$$

solving simultaneously gives

$$T = 23.5$$

dM1: Solving for T .

A1: Correct T . Accept 23.6 or AWRT 23.5

Note using $g = 9.81$ gives 23.6, also accept 23.5.

(dM1A1)

[10]

Q20.

(a) (i) $P - 500 = 2200 \times 0.8$
 $P = 1760 + 500$

M1: Equation of motion for car and caravan as a single body. Must see 2200 (or 1200 + 1000) multiplied by 0.8, and 500 (or 200 + 300). Allow sign errors.

A1: Correct equation.

M1A1

$$= 2260$$

A1: Correct value for P.

A1

(Award full marks for: (P =) 1760 + 500 = 2260 or similar to obtain correct final answer.)

OR (If finding the tension first)

$$P - 1100 - 200 = 1200 \times 0.8$$

$$P = 960 + 1100 + 200$$

M1: Equation of motion for car with their value for the tension.

Must see 1200 multiplied by 0.8, 200 and their tension.

Allow sign errors.

A1: Correct equation.

(M1A1)

$$= 2260$$

A1: Correct value for P.

(A1)

(Award full marks for: (P =) 960 + 200 + 1100 = 2260 or similar to obtain correct final answer.)

3

(ii) $T - 300 = 1000 \times 0.8$

$$T = 300 + 800$$

M1: Equation of motion for caravan.

Must see 300 and 1000 multiplied by 0.8.

Allow sign errors.

A1: Correct equation.

M1A1

$$= 1100$$

A1: Correct tension. CAO

A1

OR

$$2260 - 200 - T = 1200 \times 0.8$$

$$T = 2260 - 200 - 960$$

M1: Equation of motion for car. Must see 2260

(or candidate's P), 200 and 1200 multiplied by 0.8. Allow sign errors.

A1: Correct equation.

(M1A1)

= 1100 N

A1: Correct tension. CAO

(A1)

If candidates find tension first it must be stated in part (a)(ii) to gain any marks.

The working does not have to be repeated if seen in part (a)(i).

3

(b) (i) $15 = 7 + 0.8t$

M1: Use of a constant acceleration equation to find t, with 7, 15 and 0.8.

A1: Correct equation.

M1A1

$$t = \frac{15 - 7}{0.8} = 10 \text{ seconds}$$

A1: Correct time. CAO

A1

3

(ii) $15^2 = 7^2 + 2 \times 0.8s$

M1: Use of a constant acceleration equation to find s, with 7, 15 and 0.8.

A1: Correct equation

M1A1

$$s = \frac{15^2 - 7^2}{1.6} = 110 \text{ m}$$

A1: Correct distance. CAO

A1

OR

$$s = \frac{1}{2}(7 + 15) \times 10 = 110 \text{ m}$$

M1: Use of a constant acceleration equation to find s, with 7, 15 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)
(A1F)

OR

$$s = 7 \times 10 + \frac{1}{2} \times 0.8 \times 10^2 = 110 \text{ m}$$

M1: Use of a constant acceleration equation to find s , with 7, 0.8 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)

(A1F)

If candidates find distance first it must be stated in part (b)(ii) to gain any marks.

The working does not have to be repeated if seen in part (b)(i).

3

- (c) Resistance forces vary with speed (or velocity)

OR Speed (or velocity) changes (or increases)

OR It accelerates

B1: Correct explanation. Must not mention friction in main argument

B1

1

[13]

Q21.

- (a) $T = 2 \times 9.8 = 19.6 \text{ N}$

M1: Equating tension and weight.

A1: Correct tension CAO

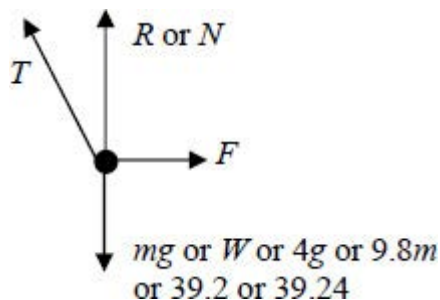
Accept $2g$

Accept 19.62 from $g = 9.81$

M1A1

2

- (b)



B1: R , F (not μR) and mg correct

B1

B1: T correct, must be in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(c) $T \cos 30^\circ + R = 4 \times 9.8$

*M1: Resolving vertically to form a three term equation.
(May be implied)*

M1

$(R =) 39.2 - 19.6 \cos 30^\circ$
 $= 39.2 - 16.9741\dots$

*A1 Correct expression for R or equation for R .
Must see $19.6 \cos 30$ or equivalent (eg $2g \sin 60$)*

A1

$= 22.2259\dots$
 $= 22.2 \text{ N (to 3sf)} \quad \text{AG}$

A1: Correct force. Must see intermediate working, for example third or fourth line of working in solution opposite.

Example: $19.6 \sin 30^\circ - R = 4 \times 9.8$ scores M1A0A0.

Use of $g = 9.81$ still gives 22.2 N as the final answer.

A1

3

(d) $T \cos 60^\circ = F$

M1: Resolving horizontally

M1

$F = 19.6 \cos 60^\circ = 9.8$

A1: Correct expression for friction

A1

$19.6 \cos 60^\circ \leq \mu(39.2 - 19.6 \cos 30^\circ)$

dM1: Use of $F = \mu R$ or $F \leq \mu R$ (do not allow $F \geq \mu R$)

dM1

$$\mu \geq \frac{19.6 \cos 60^\circ}{39.2 - 19.6 \cos 30^\circ}$$

$$\mu \geq 0.441$$

A1: Final answer of $\mu = 0.441$ or $\mu \geq 0.441$ from correct working

Use of $g = 9.81$ still gives 0.441 as the final answer.

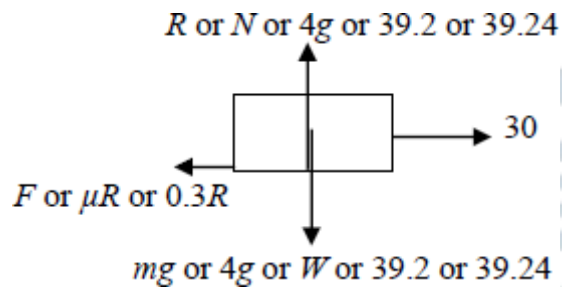
A1

4

[11]

Q22.

(a)



B1: Diagram with four forces showing arrow heads and labelled.

Ignore negative signs in labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 4 kg for the weight.

Note Accept μR for F .

B1

1

(b) $(R = 4 \times 9.8 =) 39.2 \text{ N}$

B1: Correct normal reaction. Accept $4g$

B1

1

(c) $(F =) 0.3 \times 39.2 = 11.76 = 11.8 \text{ N (to 3sf)}$

M1: Use of $(F =) \mu R$

M1

A1: Correct friction.

Accept $1.2g$ or 11.7 or 11.76 N.

Do not condone further work after the value for friction has been obtained.

A1

2

(d) $4a = 30 - 11.76$

M1: Three term equation of motion.

A1F: Correct equation.

M1A1F

$$a = \frac{30 - 11.76}{4} = 4.56 \text{ ms}^{-2}$$

A1F: Correct acceleration.

FT candidates F from part (c).

Accept 4.55 from 11.8.

A1F

3

[7]

Q23.

(a) $5g - T = 5a$

M1: Three term equation of motion with $5g$ or 49 , $5a$ (not $5ga$) and T .

A1: Correct equation.

M1A1

$$T - 3g = 3a$$

M1: Three term equation of motion with $3g$ or 29.4 , $3a$ (not $3ga$) and T .

A1: Correct equation.

M1A1

$$2g = 8a$$

$$a \left(= \frac{2g}{8} \right) = 2.45 \text{ ms}^{-2}$$

AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 2.45. If the final answer is -2.45 don't award the final A1 mark.

Special Case:

Whole String Method $2g = 8a$ and $a = \frac{2g}{8} = 2.45$
 OE M1A1A1.

5

(b) $T = 3 \times 9.8 + 3 \times 2.45$

M1: Substitution of $a = 2.45$ into a three term equation of motion to find the tension. Contains T , mg and ma where $m = 3$ or 5

M1

$= 36.75$

$= 36.8 \text{ N (to 3 sf)}$

A1: Correct tension.

Accept 36.75 or 36.7

A1

2

(c) Light and Inextensible

B1: Light

B1: Inextensible (Allow inelastic or not stretchy)

Ignore irrelevant non-contradictory assumptions.

B1B1

2

(d) (i) $0.196 = \frac{1}{2} \times 2.45 \times t^2$

M1: Use of constant acceleration equation with $s = 0.196$, $u = 0$ and $a = 2.45$ to find t .

A1: Correct equation.

M1 A1

$$t = \sqrt{\frac{2 \times 0.196}{2.45}} = 0.4 \text{ seconds}$$

A1: Correct t

A1

3

(ii) $v^2 = 0^2 + 2 \times 2.45 \times 0.196$

$v = 0.98$

M1: Use of constant acceleration equation with $s = 0.196$, $a = 2.45$, $u = 0$ and candidate's time (as needed) to find v .

A1: Correct v .

M1A1

OR

$$v = 0 + 2.45 \times 0.4 = 0.98 \text{ ms}^{-1}$$

(M1A1)

OR

$$0.96 = \frac{1}{2}(0 + v) \times 0.4$$

(M1)

$$v = 0.98 \text{ ms}^{-1}$$

(A1)

2

[14]

Q24.

(a) $P \cos 80^\circ - Q \cos 80^\circ = 250a$

M1: Horizontal equation of motion in the form

$P \cos 80^\circ \pm Q \cos 80^\circ = 250a$ or

$P \sin 80^\circ \pm Q \sin 80^\circ = 250a$

A1: Correct horizontal equation.

M1A1

$$P \sin 80^\circ + Q \sin 80^\circ = 250g$$

B1: Correct vertical equation.

Note: the above marks could be awarded for a correct vector equation.

B1

$$P - Q = \frac{250a}{\cos 80^\circ}$$

$$P + Q = \frac{250g}{\sin 80^\circ}$$

AG

$$2P = \frac{250a}{\cos 80^\circ} + \frac{250g}{\sin 80^\circ}$$

dM1: Solving for P with an attempt to eliminate Q.

dM1

$$P = 125 \left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ} \right)$$

A1: Correct result from correct working.

Must see an expression for 2P or 2P sin80°cos80°

A1

5

(b) $P \cos 80^\circ = 250a$

M1: Using $Q = 0$ into correct original equation(s) or resolving without Q .

M1

$$P \sin 80^\circ = 250g$$

$$\frac{1}{\tan 80^\circ} = \frac{a}{g}$$

dM1: Eliminating P

dM1

$$a = \frac{g}{\tan 80^\circ} = 1.73$$

A1: Correct a .

Note: use of $P = \pm Q$ scores M0dM0A0

Note: use of $P = 0$ can lead to ± 1.73 but scores M0dM0A0 unless fully justified by a symmetry argument.

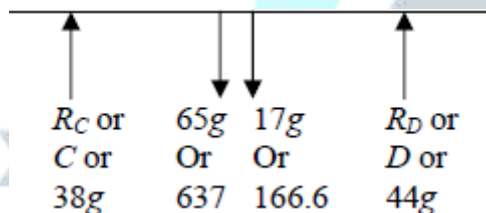
A1

3

[8]

Q25.

(a)



B1: Two weights correct and in correct relative positions.

B1: Two upward reaction forces, labelled differently.

Note all forces must be shown as arrows and have labels.

Condone use of $g = 9.81$ for calculating weights.

B1

B1

2

(b) Taking moments about C

$$3 \times 17g + 2.6 \times 65g = 44g \times d$$

B1: Seeing 2.6.

M1: Three term moment equation including 17g, 65g and 44g or 17, 65 and 44, with different distances for the 17g and 65g.

A1: Correct equation.

B1

M1
A1

$$44d = 220$$

$$d = 5$$

Distance is $5 - 4.6 = 0.4$ m

4

A1: Correct final answer.

A1

Alternative

$$R_c = 38g$$

(B1)

Taking moments about D

Could take moments about any other point

$$38g(4.6 + x) = 65g(2 + x) + 17g(1.6 + x)$$

(M1)
(A1)

$$174.8 - 130 - 27.2 = 44x$$

$$x = 0.4$$

(A1)

(4)

- (c) Gravitational force (centre of mass or weight) at mid-point (or centre) of the plank

E1

E1: Correct explanation.

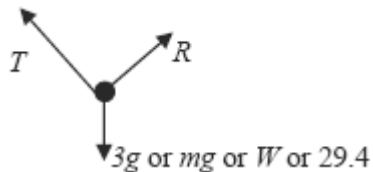
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[7]

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Q26.

(a)



B1: Correct force diagram with arrows and sensible labels.

If R is shown as vertical award B0.

If F is included, award B0

Accept a reflection of the diagram in a vertical line.

Ignore components if shown with a different notation, eg dotted lines.

B1

1

(b) $(R =) 3g \cos 60^\circ$

M1: Resolving perpendicular to the slope. Must see $\cos 60^\circ$ or $\sin 30^\circ$ or $\cos 30^\circ$ or $\sin 60^\circ$ and $3g$ or 29.4 .

NOTE: $\frac{3g}{2} = 14.7$ or equivalent without the use of a trig term scores M0.

M1

$(R =) 14.7$ **AG**

A1: Correct value from correct working. NOTE: If candidates use $g = 9.81$, deduct one mark here. If candidates obtain 14.7 from 14.715 they will have used $g = 9.81$. Note: " $R =$ " does not need to be seen.

A1

2

(c) $(T =) 3g \sin 60^\circ$

M1: Resolving parallel to the slope. Must see $\cos 60^\circ$ or $\sin 30^\circ$ or $\cos 30^\circ$ or $\sin 60^\circ$ and $3g$ or 29.4 .

M1

$(T =) 25.5$

A1: Correct value. AWRT 25.5 or truncation to 25.4. NOTE: If candidates use $g = 9.81$ again, do not penalise. Use of $g = 9.81$ gives 25.5 for the tension. Note: " $T =$ " does not need to be seen.

A1

2

[5]

Q27.

(a)



B1: Force diagram with two arrows clearly in opposite directions. Must see

300 and one other label (a letter) or 550.
Do not penalise if vertical forces included, even if wrong.

B1

1

(b) $T_1 - 300 = 500 \times 0.5$
 $(T_1 =) 300 + 250$

AG

$= 550 \text{ N}$

M1: Three term equation of motion.

M1

A1: Correct force from correct working.

Examples:

$T_1 = 300 + 250 = 550 \text{ N}$ scores M0A0

$T_1 - 300 = 250$ scores M1A1

$T_1 = 550 \text{ N}$

$T_1 = 300 + 500 \times 0.5 = 550 \text{ N}$ scores M1A1

Just $300 + 500 \times 0.5 = 550 \text{ N}$ scores M0A0

$700 + T_1 = 2500 \times 0.5$

$T_1 = 550$ scores M0A0

A1

2

(c) $T_2 - 550 - 300 = 500 \times 0.5$

M1: Four term equation of motion for Carriage 1 including 550 and 300 with mass 500

A1: Correct equation.

M1A1

$T_2 = 550 + 300 + 250 = 1100 \text{ N}$

A1: Correct force

A1

3

OR

$T_2 - 600 = 1000 \times 0.5$

M1: Three term equation of motion for Carriages 1 and 2 together including 300 twice or 600 with mass 1000.

(M1)

A1: Correct equation.

Accept $T_2 = 600 + 500$ or similar.

(A1)

$$T_2 = 600 + 500 = 1100 \text{ N}$$

A1: Correct tension

(A1)

3

(d) $P - 1100 - 400 = 2000 \times 0.5$

M1: Four term equation of motion for engine with mass 2000, a force of 400 and their answer to part (c).

M1

A1F: Correct equation.

A1F

$$P = 1100 + 400 + 1000 = 2500$$

A1F: Correct force.

A1F

3

OR

$$P - 1000 = 3000 \times 0.5$$

M1: Three term equation of motion for whole train with mass 3000 and 1000 (OE) force.

(M1)

A1F: Correct equation.

(A1F)

$$P = 1000 + 1500 = 2500$$

A1F: Correct force

*Follow through from incorrect T_2 in part (c).
Don't penalise candidates who use a letter other than P .*

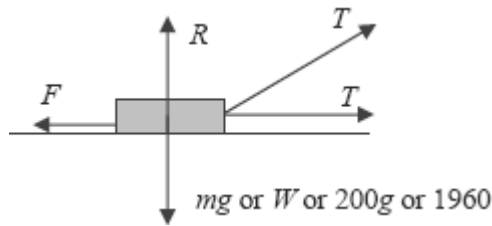
(A1F)

(3)

[9]

Q28.

(a)



B1: F , R and mg (or equivalent) with arrows and labels.

B1

*B1: Two **equal** tension forces with arrows and labels.*

B1

Ignore components if shown with a different notation, eg dotted lines.

2

(b) $R + T \sin 20^\circ = 1960$

OR

$R + T \sin 20^\circ = 200g$

M1: Resolving vertically with three terms. Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and $200g$ or 1960 .

A1: Correct equation.

M1A1

$(R =) 1960 - T \sin 20^\circ$

OR

$(R =) 200g - T \sin 20^\circ$

A1: Correct expression for the normal reaction.

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 1962 instead of 1960

A1

3

(c) $T \cos 20^\circ + T - F = 200 \times 0.3$

M1: Four term equation of motion. Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and a second T term with no trig.

A1: Correct equation

M1A1

$T \cos 20^\circ + T - 0.4 (1960 - T \sin 20^\circ) = 200 \times 0.3$

M1: Use of friction law with their expression for R , provided that R has two terms. Note that this mark does not depend on any previous marks.

Example:

If Candidate gives 1960 as answer to part (b),
 then: $F = 0.4 \times 1960 = 784$ scores M0 here

M1

$$T = \frac{60 + 784}{\cos 20^\circ + 1 + 0.4 \sin 20^\circ} = 406$$

dM1: Solving for T .

dM1

Note: This mark requires both of the
 previous M marks.

A1: Correct tension.

Accept AFWW 406 to 407.

Note: If $g = 9.81$ is used should get 407
 instead of 406.

A1

5

[10]

Q29.

(a)



B1 for four forces

B2 for two different reactions and 30g and
 20g marked

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B2

2

(b) Taking moments about A:

$$3.2 \times 30g = R_B \times 5$$

B1 for 3.2

M1B1

$$R_B = 19.2g$$

AG

A1

3

(c) Resolve vertically: $R_A + R_B = 50g$

Can be awarded in (b)

M1

$$R_{\perp} = 30.8g \text{ or } 302 \text{ N}$$

A1

2

- (d) Gravitational force acts through mid-point of the rod

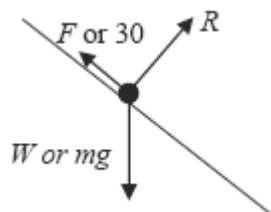
E1

1

[8]

Q30.

(a)



B1: Correct force diagram with arrows and labels.

B1

1

- (b) $30 = mg \sin 42^\circ$

M1: Two term equation by resolving parallel to the slope. Must include g.

A1: Correct equation.

M1A1

$$m = \frac{30}{9.8 \sin 42^\circ} = 4.57$$

A1: Correct value for m.

A1

3

- (c) $(R =) mg \cos 42^\circ = 33.3 \text{ N}$

M1: Resolving perpendicular to the slope to find R. Must include g.

A1: Correct value.

M1A1F

2

- (d) $30 = \mu R$

$$\mu = \frac{30}{R} = 0.900$$

M1: Use of $F = \mu R$

M1

M1: Solving for μ .

M1

A1F: Correct value for μ . Accept 0.9.

A1F

3

Use of $g = 9.81$

(b) 4.57 B1

(c) 33.3 B1

(b) 0.900 B1

[9]

Q31.

(a) $12g - T = 12a$

M1: Three term equation of motion, with $12g$ (or 117.6), $12a$ (not $12ga$) and T .

A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T .

A1: Correct equation

M1A1

$4g = 20a$

$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2} \quad \text{AG}$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g = 20a =$ and

$$a = \frac{4g}{20} = 1.96 \quad \text{OE M1A1A1}$$

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$$v = 9.68 \text{ m s}^{-1}$$

*A1F: Correct v . FT initial velocity from (c)(i).
For example 11.8 from 7.84.*

A1F

3

Use of $g = 9.81$

(a) 1.962

M1A1M1A1A0

(b) 94.2

M1A1

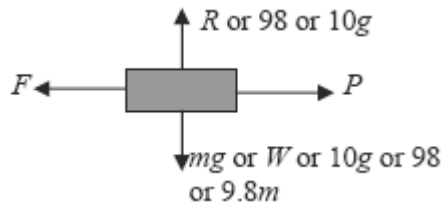
(c) (ii) 9.69

M1A1A1

[12]

Q32.

(a)



B1: Correct force diagram with arrows and labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 10 kg for the weight.

Note: Do not accept μR or $0.5R$ for F .

B1

1

(b) (i) $(R = 10 \times 9.8 =) 98 \text{ N}$

B1: Correct normal reaction. Accept 10g.

No need to see the letter R or working.

B1

1

(ii) $(F \leq) 0.5 \times 98$

$(F \leq) 49$

B1: Correct maximum value for friction.

Accept 5g.

No need to see the letter F or any working. Ignore any inequalities.

For FT, must be 0.5 of candidate's answer to (b)(i).

B1F

1

(iii) $(F =) 30 \text{ N}$

B1: Correct friction. Allow – 30.

B1

1

(c) $80 - 49 = 10a$

M1: Three term equation motion, containing 80, candidate's 49 and $10a$ (not $10ga$) in any combination.

A1F: Correct equation including signs.

M1A1F

$a = 3.1 \text{ m s}^{-2}$

A1F: Correct acceleration.

FT candidate's answer to (b)(ii).

A1F

3

Allow use of $g = 9.81$

- (b) (i) 98.1 B1
 (b) (ii) 49.05 or 49.1 or 49 B1
 (c) 3.095 or 3.09 or 3.1 M1A1A1

[7]

Q33.

(a) $12g - T = 12a$

M1: Three term equation of motion, with $12g$ (or 117.6), $12a$ (not $12ga$) and T .

A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T .

A1: Correct equation

M1A1

$4g = 20a$

$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.

A1

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Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is -1.96 don't award final A1 mark.
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Special Case:

Whole String Method $4g = 20a$ and

$a = \frac{4g}{20} = 1.96$ OE M1A1A1

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$v = 9.68 \text{ m s}^{-1}$

A1F: Correct v . FT initial velocity from (c)(i). For example 11.8 from 7.84.

A1F

3

(iii) $4 = \frac{1}{2}(-3.92 + 9.68)t$

M1: Use of $s = \frac{1}{2}(u + v)t$

A1: Correct values.

M1

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A1

A1: Correct signs.

A1

$t = 1.39$

dM1: Solving for t .

dM1

A1: Correct t .

A1

OR

$$\begin{aligned}
 -4 &= 3.92t - 4.9t^2 \\
 4.9t^2 - 3.92t - 4 &= 0 \\
 t &= \frac{3.92 \pm \sqrt{3.92^2 - 4 \times 4.9 \times (-4)}}{2 \times 4.9}
 \end{aligned}$$

M1: Forming a quadratic with candidates u from (c)(i) or v from (c)(ii) with 4.9 or 9.8.

(M1)

A1: Correct terms in quadratic.

(A1)

A1: Correct signs in quadratic.

(A1)

$$t = 1.39 \text{ or } t = -0.588$$

dM1: Solving quadratic (do not penalise for negative discriminant).

(dM1)

$$t = 1.39$$

A1: Correct root seen (other root does not need to be seen).

(A1)

OR

$$t_{\text{up}} + t_{\text{down}} = 0.4 + 0.4 + 0.588$$

M1: Finding total time from two or three times.

(M1)

A1: 0.4 or 0.8 seen.

(A1)

dM1: Finding second or third time for downward motion.

(dM1)

A1: Obtaining 0.588 or 0.988.

(A1)

$$= 1.39 \text{ (to 3SF)}$$

A1: 1.39. Accept 1.38.

(A1)

OR

$$9.68 = -3.92 + 9.8t$$

M1: Use of $v = u + at$

(M1)

A1: Correct values.

(A1)

A1: Correct signs.

(A1)

$$t = \frac{13.6}{9.8} = 1.39$$

dM1: Solving for t

(dM1)

A1: Correct t

(A1)

5

Use of $g = 9.81$

(a) 1.962 M1A1M1A1A0

(b) 94.2 M1A1

(c) (ii) 9.69 M1A1A1

(c) (iii) 1.39 M1A1A1dM1A1

[17]

Q34.

(a) $11g - T = 11a$

M1: Equation of motion for A, containing T , $11g$ or 107.8 and $11a$.

M1

A1: Correct equation

A1

$$T - 9g = 9a$$

M1: Equation of motion for B containing T , $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$$2g = 20a$$

AG

$$a = 0.98 \text{ m s}^{-2}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and $a = 2g / 20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(b) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225 \text{ m}$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$$0.49^2 = 0^2 + 2 \times 0.98s$$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v .

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of $d/2$ in

their original equation.

M1

$$= 0.245 \text{ m}$$

A1: Correct final distance. Allow 0.244 or 0.246.
(Use of $0.5 \times 0.49 = 0.245$ scores zero unless justified)

A1

4

[11]

Q35.

(a) $4a = 4g \sin 40^\circ$

M1: Resolving and application of Newton's second law. Allow $\cos 40^\circ$

M1

A1: Correct expression.

A1

$$a = g \sin 40^\circ = 6.30 \text{ m s}^{-2} \quad \text{AG}$$

A1: Correct result from correct working.
Must see 6.30 not 6.3.
Just seeing $g \sin 40^\circ = 6.30 \text{ m s}^{-2}$ scores full marks.
Use of $g = 9.81$ gives 6.31, M1A1A0, but don't penalise again on the same script.

A1

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3

(b) $0.9 = 0 + \frac{1}{2} \times a \times 0.6^2$

M1: Use of a constant acceleration equation to find a , with $s = 0.9$, $u = 0$ and $t = 0.6$.

M1

A1: Correct equation

A1

$$a = \frac{0.9 \times 2}{0.6^2} = 5 \text{ m s}^{-2}$$

A1: Correct acceleration

A1

ALT Method

$$09.9 = \frac{1}{2}(0 + v) \times 0.6$$

$$v = 3$$

No marks at this stage.

$$3 = 0 + 0.6a$$

*M1: Constant acceleration equation with
 $u = 0$ and $t = 0.6$.*

(M1A1)

A1: Correct equation

$$a = 5 \text{ m s}^{-2}$$

A1: Correct acceleration.

(A1)

3

- (c) The acceleration is reduced because of air resistance or the fact that there is friction.

B1: Must mention air resistance/resistive forces or friction. Do not allow air friction.

B1

1

[7]